Function-Oriented Software Design

Solved and Practice Exercises

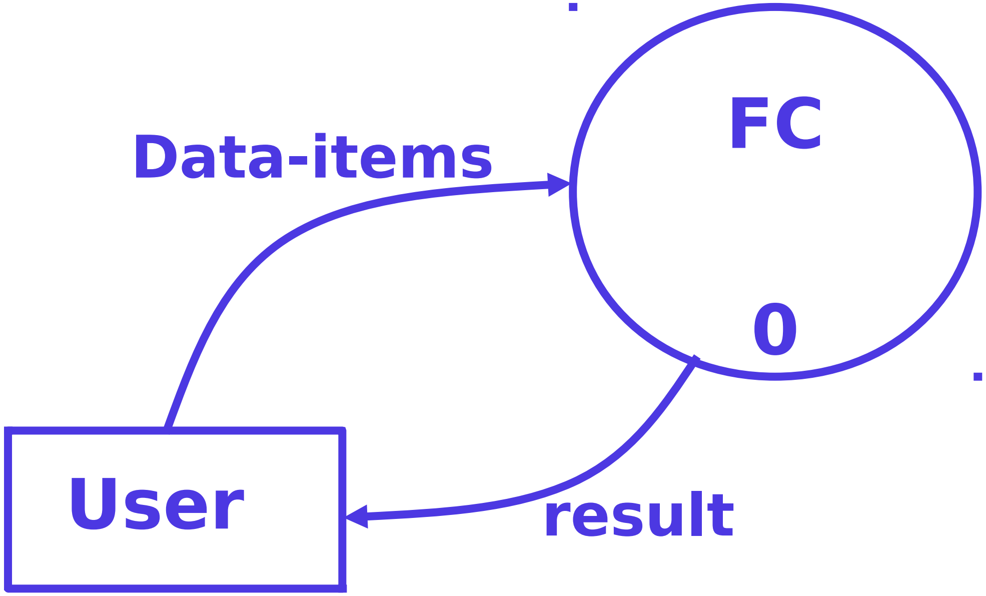
Formulae Calculator (FC) Software

- Consider a software called formulae calculator software
  - Reads three integers in the range of -1000 and +1000
  - Finds out the root mean square of the three input numbers
  - Displays the result.
- The context diagram is simple to develop:
  - The system accepts 3 integers from the user
  - Returns the result to him.
- From a cursory analysis of the problem description:
  - We can see that the system needs to perform several things.
    - Accept input numbers from the user:
    - Validate the numbers,
    - Calculate the root mean square of the input numbers
    - Display the result.

Structured Analysis (SA)

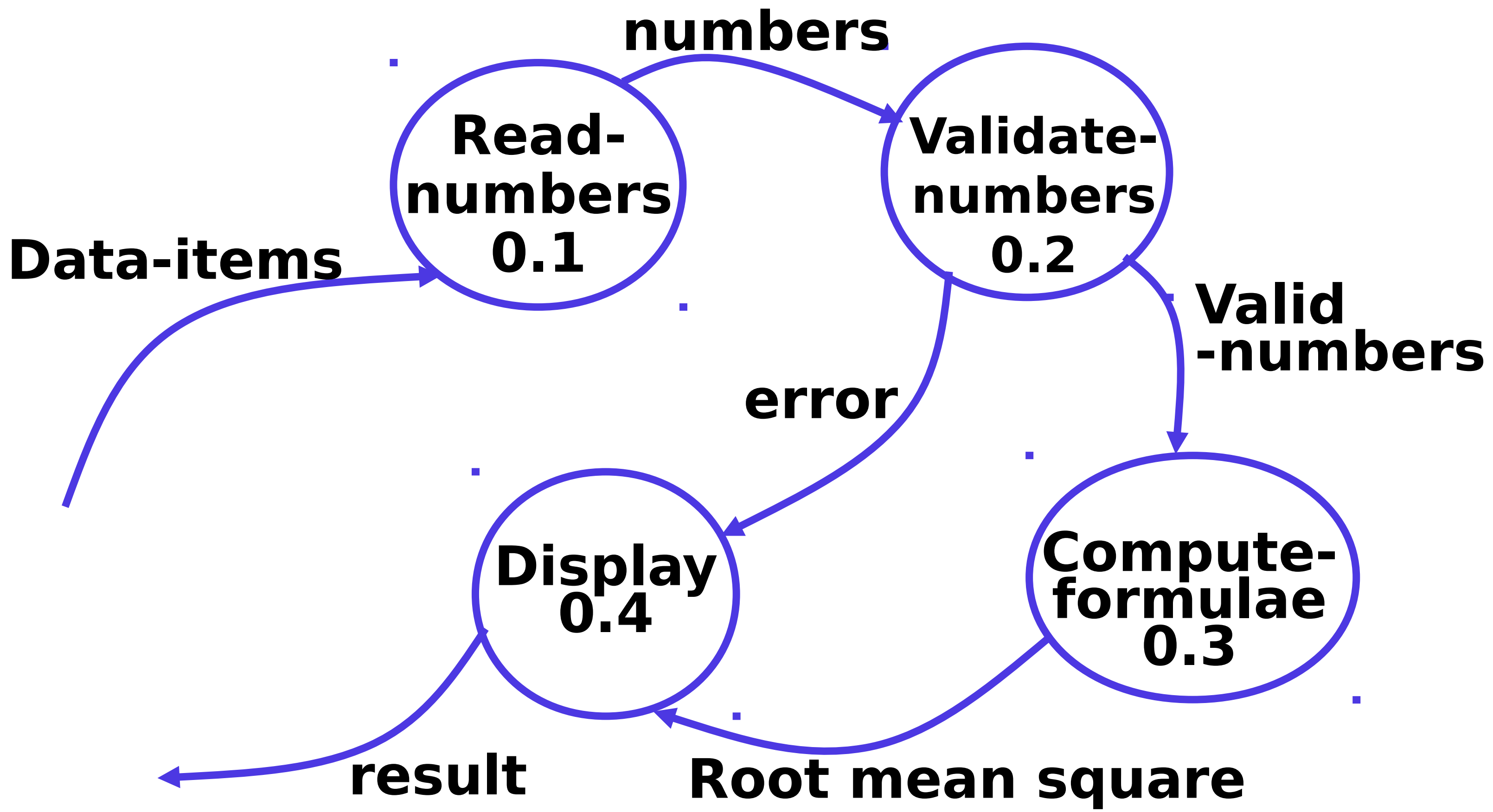
Input: Requirements Doc / SRS
Output: Different DFDs

Formulae Calculator (FC) Software : Context Diagram



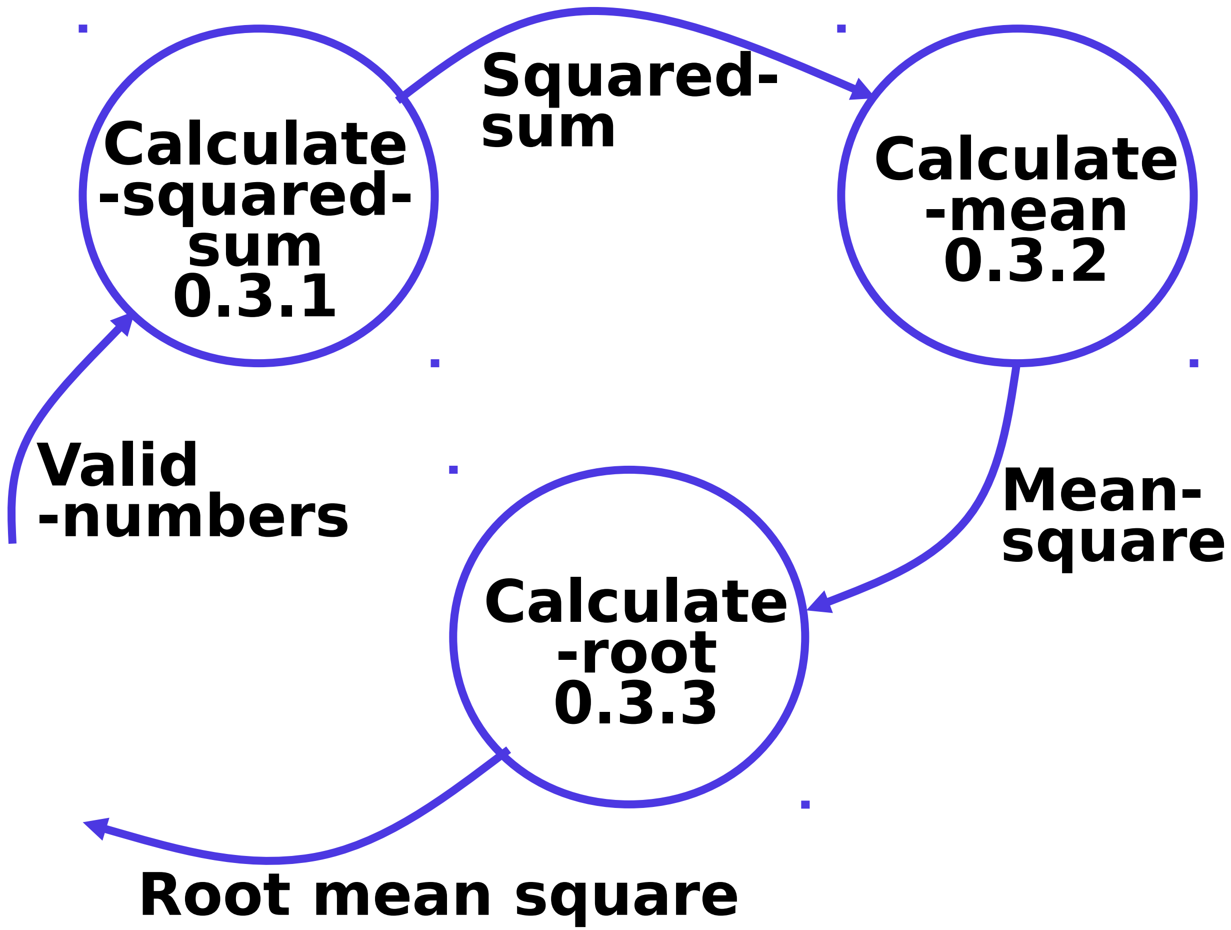
Level 0 DFD

Formulae Calculator (FC) Software : Level 1 DFD



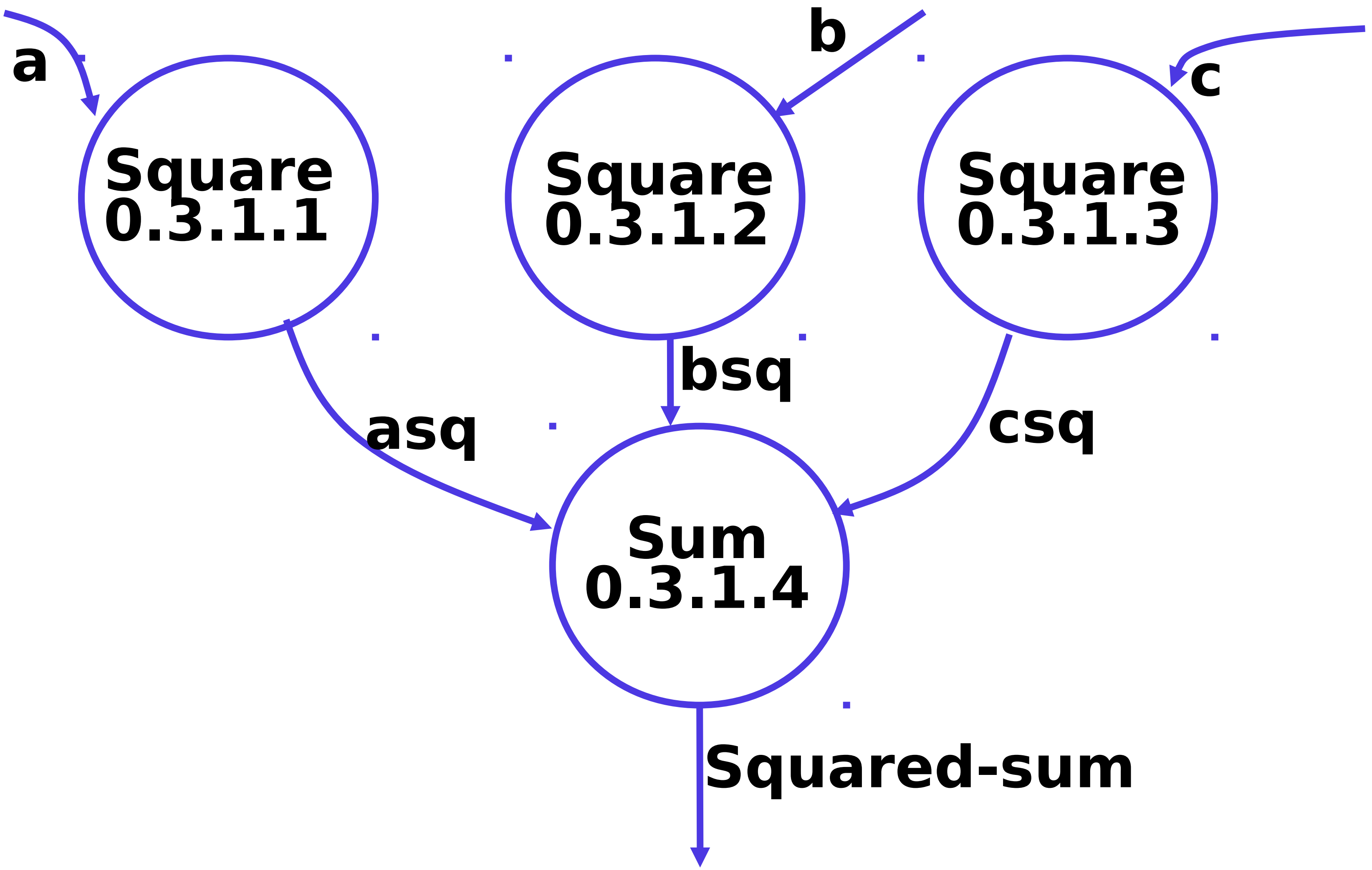
Note: 0.1, 0.2, 0.4 are operations that no more require decomposition.
However, 0.3 can be further decomposed. Hence Level 2 DFD for 0.3 in next slide.

Formulae Calculator (FC) Software : Level 2 DFD



Note: 0.3.1 can be further decomposed. Hence Level 3 DFD for 0.3.1 in next slide.

Formulae Calculator (FC) Software :
Level 3 DFD (for 0.3.1 in level 2 DFD)

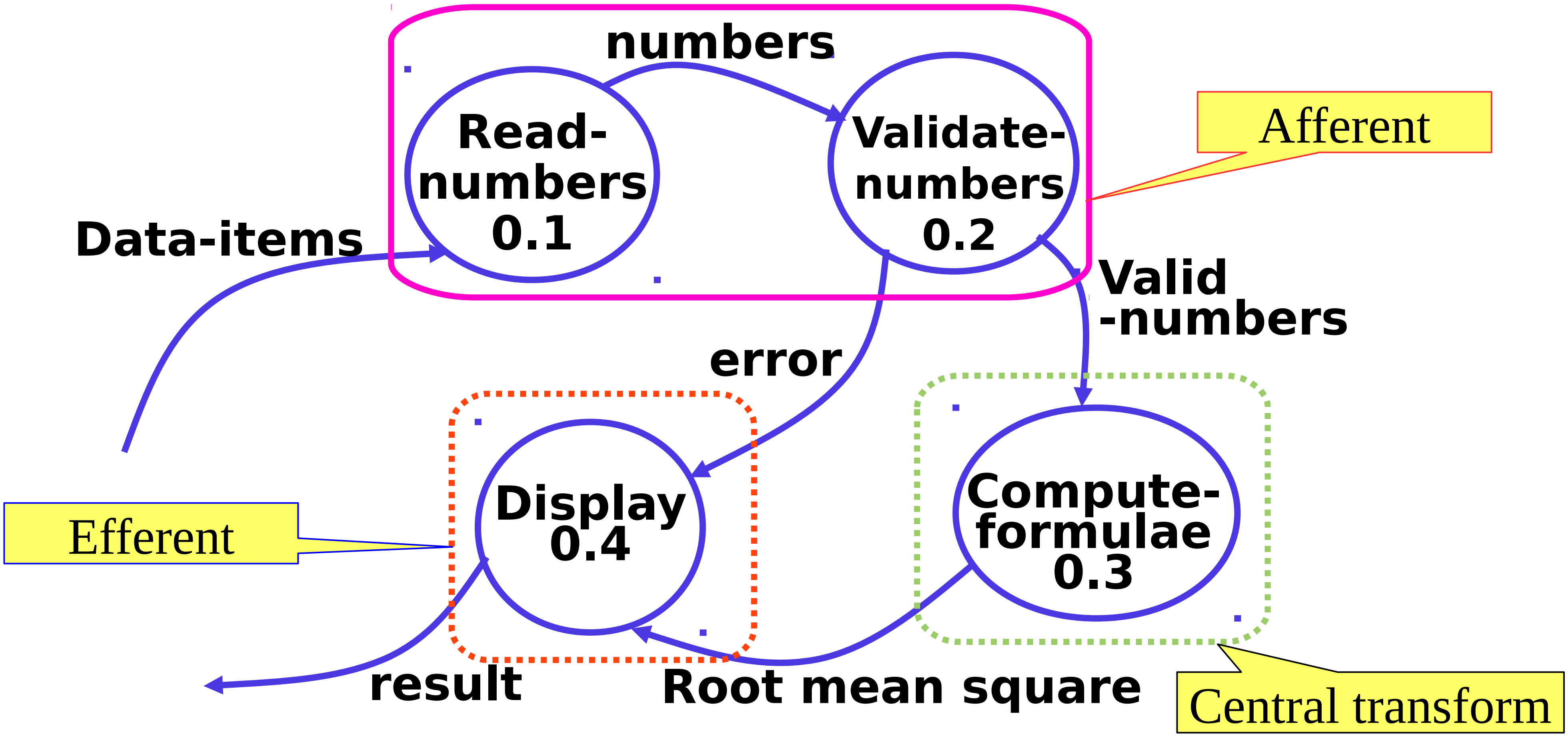


Note: You may add Level 3 DFD for 0.3.2 and 0.3.3, if needed.

Structured Design (SD)

Input: DFD obtained in SA phase
Output: Module Structure Chart and Data Dictionary

Formulae Calculator (FC) Software : Process of Structured Design



- By observing the level 1 DFD:
  - Identify 0.1 and 0.2 bubbles as the afferent branch – Input section
  - Display 0.4 as the efferent branch – output section
  - Compute-formulae 0.3 is the remaining portion as the central transform – logic processing

Formulae Calculator (FC) Software : Module Structure Chart